

ECON 695 Problem Set 5

October 26, 2025

Due: 11:59PM Central Time on Monday 11/03/2025.

Note: I provide a Jupyter notebook (**PS5_twins.ipynb**) to guide the analysis. If you use AI assistance, make sure you have an AI disclaimer that indicates where/how you use AI tools. Grader will take 2 points off if she recognizes AI-generated code but does not see a disclaimer.

Twins Study

In this problem set we will fit a few models to the Princeton Twins Survey data. The data set is called **twins.csv**. The variables are:

- famid = family id variable
- t=1,2 for twin #1 or twin #2
- age = age (some observations have coded in part year values)
- educ = education
- oeduc = education of twin
- lw = log wage
- married = dummy 1 if married 0 if not
- omarried = dummy if twin is married
- female = 1 if female
- ofemale = 1 if twin is female.
- exp = “labor market experience” = age - educ - 6
- oexp = experience of twin

NOTE: all twins are one of two identical twins in the data set. So female=ofemale in all cases.

1. Estimate a simple model relating log wages to: education, experience, experience-squared, married status, and female.
2. Fit the same model separately for men and women. Does the model look different?
3. Construct the mean family marriage rate for each person in the data set (i.e., the fraction of the twins that is married, which can be 0, 1/2, or 1).

- (a) Verify that when you regress marriage of a twin on the average fraction of the siblings who are married, you get a coefficient of 1. Hint: we proved this in lecture 8. Here you are asked to run this “auxiliary” regression and check if the regression coefficient is indeed 1.
 - (b) Add mean fraction of siblings married to your gender-specific wage models from part 2. How does the addition of this variable affect the estimated coefficient on marriage. Give an interpretation of the patterns and how they differ between men and women.
 - (c) Instead of controlling for the mean family marriage rate, estimate the gender-specific wage models from part 2 with a control for the other twin’s marriage status (variable “omarried”). Compare the regression coefficients on (married, omarried) with the coefficient on (married, mean married) in (b).
4. (Three ways to get the within estimator) In lecture 8 we show three ways to get the within estimator - de-meaned relative to the mean, fixed effects, and control function. Let’s run 3 regressions for **male** twins.
- (a) De-mean variables relative to the mean in each family: replace x_{it} by $x_{it} - \bar{x}_i$ where x_{it} includes education, experience-squared, and married status, and replace y_{it} by $y_{it} - \bar{y}_i$. Fit a regression of $(y_{it} - \bar{y}_i)$ on $x_{it} - \bar{x}_i$.
 - Note we dropped experience from the regression. See how experience is defined. The de-meaned experience = - demeaned education (correlation = -1).
 - (b) Fit a regression of y_{it} on x_{it} and family fixed effects.
 - (c) Fit a regression of y_{it} on x_{it} and means $\bar{x}_i$ in each family. Verify if the coefficients on x_{it} are the same as in (a) and (b).